Procurement, hybrid build: my track on Nhien's Rev B frame

Date 15 Oct 2026. Prices S$ ex GST as seen in listings; confirm at checkout. Rev B = Nhien's pack (`RB4107 Rev B`), whose BOM covers everything in section 2.

1. Mine (track side)

Part	Qty	Make / buy	Where	Price	Arrives
Track, TPU 95A, `stl/track_TPU_x4.stl`	4	**Make**: Fab Lab UP300 (TPU head)	Element14 SG TPU 95A filament if the lab has none	~S$45/kg, uses 130 g	filament 1-2 days; 3 h print each
Fit-test slice, `stl/fit_test_track_5mm_TPU.stl`	1	Make	same	-	10 min
Stand-in pulley, `stl/stand_in_pulley_20T_PLA_test_only.stl`	1	Make, PLA	any printer	-	1 h
Timing pulley HTD 5M 20T, 15 mm belt, flanged, **plain (pilot) bore**	8	**Buy**	MISUMI SG, Timing Pulleys HTD 5M	~S$20 each	ships 1 day from SG stock; allow 2-5 days
Pulley bore modification: 4 x D-bore Ø8 (M1), 4 x Ø16 H7 (M2)	8	**Make**: Professional Workshop request, Rev B sheet 21	E3-01-06	free	workshop queue, allow 1-2 weeks

2. Nhien's (frame, drive, hinge, electronics), from the Rev B BOM

Part	Qty	Make / buy	Where
Side plates P1 (front) and P2 (rear), 5 mm Al, laser + tap	2 + 2	Make	DXF in the pack; Professional Workshop or a laser shop on the vendor list
Slide rails P3-P6, UHMW strip 189 x 20 x 5	8	Buy + drill	MISUMI SG UPA-189-20-5
Rail brackets RS PRO 427-991 (angle 15 x 14.6 x 10)	32	Buy	RS SG stock no. 427-991
Hinge shafts P20 A 82.3 + B 77.3 (or one 212), Ø8 D-shaft flat 7	2	Buy stock, cut	RS SG linear shafts or MISUMI SG
Bearing 688-2RS 8 x 16 x 5	14	Buy	RS SG ball bearings
Shoulder screw ISO 7379 Ø8 x 25 M6 + washers, shim, spacer, nyloc	4 sets	Buy	RS SG shoulder screws
2020 profile x 110 (P7) x2, 3030 x 160 (P8) x2, corner posts P15 x4	1 m + 0.5 m	Buy, cut	MISUMI SG / local
Corner brackets P18 (25x25x4 angle) x4, P19 (20x20x3) x4, T-nuts, E-clips, shims	-	Buy / make	Rev B BOM sections 3-4

Deck P12, tray P16, cover P22, motors, drivers, ESP32, battery	-	Theirs	Rev B BOM

3. What is no longer needed from my earlier list

Printed side plates (4), printed spacers (6), M4 rods and nuts, Ø8 plain rod for three axles, 608ZZ x10, M4 set screws. Keep the STL files in case the team falls back to the all-printed frame.

4. Order of events

1. Order the 8 plain-bore pulleys today (critical path, then the workshop bore mod); print the stand-in pulley and fit-test slice; print the 4 tracks as soon as a UP300 slot is free.
2. Nhien: laser plates, order rails, bearings, shoulder screws, profiles.
3. Workshop: pulley bores, plate tapping, shaft cutting (one request form).
4. Assemble per Rev B sheet 3; fit my tracks on the real pulleys; tension at the slots.